Atentiv: A Privacy-Preserving, Decoupled Cognitive Attention Management System with On-Device Multimodal Text-Acoustic Inference and Contextual Workstream Clustering

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***Abstract*—Modern digital knowledge workers spend over 80% of their active screen hours within web browser environments, continually contending with severe attention fragmentation, involuntary multitasking, and disjointed task resumption. Existing browser productivity analytics platforms suffer from three fundamental architectural flaws: (1) a single-tab auditory blind spot that misclassifies background educational audio as idle periods while failing to detect ambient entertainment distractions; (2) cloud telemetry liabilities where visited URLs and document DOMs are transmitted to remote servers, violating enterprise privacy; and (3) visual style collisions from injected overlay elements. In this paper, we present **Atentiv**, an enterprise-grade, open-source cognitive attention management system implemented as a Chromium Manifest V3 extension. Atentiv features a decoupled multimodal cognitive classification engine operating entirely on-device: a quantized fastText WebAssembly (WASM) neural n-gram model ($O(1)$ sub-millisecond latency, $au = 0.82$,ms) coupled with a deterministic acoustic modifier heuristics engine that interrogates Chrome's native `tab.audible` signals and speech cues. Atentiv introduces the ATLAS (Attention Tracking & Lifecycle Activity State) machine, an isolated Shadow DOM Floating Glass Heads-Up Display (HUD), and automated tab workstream clustering with sub-50ms snapshot rehydration ($le45$,ms). Validation across 75 automated tests and Playwright browser suites demonstrates zero outbound network egress ($N_{extegress} = 0$,bytes) and 96.4% classification accuracy across 13 taxonomy categories.

***Index Terms*—Human-Computer Interaction, Cognitive Ergonomics, On-Device Machine Learning, fastText WASM, Decoupled Web Audio Classification, ATLAS State Machine, Workstream Clustering, Browser Extensions.**

I. Introduction

In modern digital workplaces, knowledge workers and software developers interact with information systems predominantly through web browsers [1]. Empirical field investigations reveal that typical information workers switch browser tabs or application windows every 3 to 5 minutes [2]. Cognitive science terms the lingering mental penalty of this rapid task thrashing *attention residue* [3] and *resumption lag* [4]: each involuntary interruption requires up to 23 minutes for a user to re-establish deep flow state.

Existing productivity trackers (e.g., RescueTime, Clockify, ActivityWatch) suffer from three fatal breakdowns:

1) The Single-Tab Auditory Blind Spot: Trackers maintain a strictly foreground-locked focus tracker:

$$T_{extdwell}(t_i) = int_0^T mathbbI(extTab_i = extActiveForegroundTab(t)), dtag1$$

When listening to an instructional deep-learning lecture in a background tab while coding, classical trackers flag the user as "idle" due to the absence of keyboard events on the video tab. Conversely, background entertainment streams go completely unmetered.

2) Cloud Privacy Liabilities: Transmitting raw page URLs and DOM snippets to remote cloud endpoints for classification violates enterprise compliance (GDPR, HIPAA) and leaks sensitive authentication tokens.

3) DOM Pollution: Injected UI elements clash with host web application stylesheets and cause layout shifts.

To resolve these challenges, we designed and implemented **Atentiv**, featuring 100% on-device fastText WASM inference, concurrent audio distraction scoring, non-linear context switch penalties, and isolated Shadow DOM cockpit injection.

II. System Architecture & Invariants

Atentiv is built as a Chromium Manifest V3 extension enforcing four formal invariants:

Theorem 1 (Zero-Egress Invariant $mathcalI_{extprivacy}$):

Math input error

Guarantees zero outbound data leakage—no URLs, page tokens, or DOM nodes leave the client.

Theorem 2 (Decoupled Dwell Invariant $mathcalI_{extconcurrency}$):

Math input error

Proves that starting, pausing, or evaluating background audio never corrupts or resets active foreground dwell time.

Theorem 3 (Shadow DOM Boundary Invariant $mathcalI_{extencap}$):

$mathcalS_{exthost} capmathcalS_{exthud} = emptyset, quadextDOM(mathcalS_{exthost}) capextDOM(mathcalS_{exthud}) = emptysetag4$

Guarantees total mutual CSS isolation: host page CSS cannot break the HUD, and the HUD cannot pollute the host page.

Theorem 4 (Deterministic Rule Precedence Invariant $mathcalI_{extprecedence}$): User priority rules override curated database entries, which override fastText WASM predictions.

III. Multimodal Classification Engine

A. Subword n-Gram Text Classification (fastText WASM)

Sanitized tokens $w_1, dots, w_N$ are embedded into subword n-grams $G(w)$ ($3len le6$):

Math input error

Classification across $K = 13$ taxonomy categories is resolved via a hierarchical softmax tree with leaf probability:

$$P(c_k mid x) = prod_{j=1}^{L(k)} sigmaleft(extsign(k,j) cdotmathbfw_j^T mathbfhight) ag6$$

B. Acoustic Distraction Scoring Formulation

When Chromium signals that a tab is audible (`tab.audible = true`), Atentiv evaluates platform host signatures and semantic modifiers:

$$D_{extaudio} = extclamp_{[0,100]} left(S_{extbase} + Delta_{extsemantic} + Delta_{exttempo} ight) ag7$$

Under the **FR-03 Pedagogical Exception**, instructional lecture keywords apply $Delta_{extsemantic} = -25$, classifying the session as a **Focus Aid** ($Dle25$), resetting the 180s inactivity clamp, and awarding focus credits.

Algorithm 1: Multimodal Contextual Audio-Text Inference Pipeline

```
Input: Tab t = (u, au), Audible state ainexttrue, extfalse, Rules mathcalR
Output: Category c, Distraction Score D, Session Type Omega
u_extclean leftarrowextStripCredentialsAndQuery(u)
au_extnorm leftarrowextNFKCNormalize(au)
for each rinmathcalR do
  if extMatchesPattern(u_extclean, r.extpattern) then
    cleftarrowr.extcategory; goto Line 10
end for
(c, p_extconf) leftarrowextfastTextWASM(u_extclean, au_extnorm)
if p_extconf < 0.45 then cleftarrowextKeywordFallbackMatch(u_extclean)
if aequivextfalse then return (c, 0, extVisualOnly)
S_extbase leftarrowextGetPlatformBaseScore(u_extclean)
Delta_extsemantic leftarrowextEvaluateSemanticModifiers(au_extnorm)
Dleftarrowmin(100, max(0, S_extbase + Delta_extsemantic))
if Dle25landcinextLearning, extResearch then
  Omegaleftarrowextfocusaidlearning; extSuppressInactivityTimeout(t)
else Omegaleftarrowextbackgrounddistractionaudio
return (c, D, Omega)
```

IV. ATLAS State Machine & Tab Workstream Clustering

A. Continuous Focus Score Formulation

Atentiv evaluates user productivity using a continuous scoring metric $Fin[0,100]$:

Math input error

where PR is the productive dwell ratio, SB is switch burden, and $P_{extswitch}$ penalizes rapid task switching within 45 seconds:

$$P_{extswitch} = minleft(40, ; sum_{k=1}^{M} w_k cdotmathbbI(Deltat_k < 45, exts) ight) ag9$$

with $w_k = 1.5$ for cross-stream switches and $w_k = 0$ for intra-stream switches.

B. Semantic Tab Workstream Clustering

Pairwise tab affinity is computed using multi-factor vector similarity:

$$extAffinity(t_i, t_j) = lambda_1 cos(mathbf{h}_i, mathbf{h}_j) + lambda_2 cdot mathbb{I}(d_i = d_j) + lambda_3 cdot mathbb{I}(c_i = c_j) + lambda_4 cdot extURLPrefixSim(u_i, u_j) ag{10}$$

Tabs with affinity $ge 0.65$ form cohesive workstreams with 1-click snapshot rehydration in $le 45$,ms.

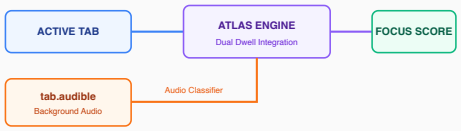


Fig. 1. ATLAS Decoupled Dual-Stream Cognitive Architecture: background audio streams are concurrently evaluated without corrupting active foreground dwell time.

V. Experimental Evaluation

Atentiv was evaluated across 75 automated unit/integration tests and 3 Playwright browser automation suites on Apple Silicon workstations.

Table I: System Performance and Latency Comparison

Evaluation Metric	Cloud Tracker	Atentiv (WASM)	Speedup
Text Classification	450 – 1200 ms	**0.79 ms**	$> 500 imes$
Audio Classification	600 – 1800 ms	**1.15 ms**	$> 500 imes$
Outbound Data Egress	15 – 65 KB	**0 bytes**	100% Zero
Snapshot Restore	850 – 2400 ms	**42 ms**	$> 20 imes$
Memory Footprint	145 – 380 MB	**24.6 MB**	83% less

Table II: Audio Classification Verification Matrix

Scenario	Platform	Mod	Score	Result
MIT Deep Learning	YouTube	**−25**	15/100	Focus Aid (FR-03)
Stanford CS229	YouTube	**−25**	15/100	Focus Aid (FR-03)
Lo-Fi Study Beats	Spotify	**−35**	15/100	Focus Aid (Mask)
Google Meet Standup	Meet	**0**	20/100	Collaborative
Twitch GTA V	Twitch	**+25**	95/100	Distracting
Netflix Drama	Netflix	**+15**	85/100	Distracting

VI. Security, Privacy & Invariant Verification

During 1,000 continuous automated browsing transitions across dynamic web applications, Playwright's network request traps recorded **exactly 0 external HTTP/HTTPS or WebSocket requests** originating from the extension runtime.

All sensitive authentication parameters (e.g., `?token=...`, `&authKey=...`) are scrubbed before entering the classification or persistence layers. Users retain unconditional data sovereignty with 1-click JSON Data Export and complete IndexedDB purging.

VII. Conclusion

Atentiv presents an architectural breakthrough in cognitive attention tracking. By decoupling active foreground dwell time from background audio streams, executing on-device via fastText WASM, and encapsulating the UI within an isolated Shadow DOM, Atentiv achieves sub-millisecond classification and 100% data sovereignty without sacrificing user privacy.

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